

Large Language Models for Disease Diagnosis: A Scoping Review

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Abstract. Automatic disease diagnosis has become increasingly valuable in clinical practice. The advent of large language models (LLMs) has catalyzed a paradigm shift in artificial intelligence, with growing evidence supporting the efficacy of LLMs in diagnostic tasks. Despite the increasing attention in this field, a holistic view is still lacking. Many critical aspects remain unclear, such as the diseases and clinical data to which LLMs have been applied, the LLM techniques employed, and the evaluation methods used. In this article, we perform a comprehensive review of LLM-based methods for disease diagnosis. Our review examines the existing literature across various dimensions, including disease types and associated clinical specialties, clinical data, LLM techniques, and evaluation methods. Additionally, we offer recommendations for applying and evaluating LLMs for diagnostic tasks. Furthermore, we assess the limitations of current research and discuss future directions. To our knowledge, this is the first comprehensive review for LLM-based disease diagnosis.

Introduction

Automatic disease diagnosis is a crucial task in clinical scenarios that takes clinical data as input, analyzes patterns, and generates potential diagnoses with minimal or no human intervention¹. Its

significance in healthcare is multifaceted. First, it enhances diagnostic accuracy, supports physicians in clinical decision-making, and addresses disparities in healthcare access by providing more high-quality diagnostic services². Second, automatic diagnosis improves the efficiency of healthcare professionals^{3,4}, which is particularly valuable for clinicians managing larger panels of patients with increasing age and multiple morbidities⁵. For instance, DXplain⁶ was a diagnostic system that utilized patients' signs, symptoms, and laboratory data to generate a list of potential diagnoses, along with a justification for why each condition should be considered. Additionally, online services further facilitate early diagnosis or large-scale screening of certain diseases^{4,7}, such as mental health disorders, by raising awareness in the early stages and helping to prevent potential risks. For example, several studies investigated using social media posts for large-scale depression identification⁸ and suicide risk prediction⁹.

Recent advancements in artificial intelligence (AI) have driven the development of automated diagnostic systems through two stages^{10–13}. Initially, machine learning techniques such as support vector machines and decision trees were employed for disease classification^{14,15}, which typically involved four steps: data processing, feature extraction, model optimization, and disease prediction. With larger datasets and sufficient computational power, deep learning methods later dominated the development of diagnostic tasks^{2,16}. These approaches leveraged deep neural networks (DNNs), including convolutional neural networks^{1,17}, recurrent neural networks¹⁸, and generative adversarial networks¹⁹, enabling end-to-end feature extraction and model training. For example, a convolutional DNN with 34 layers achieved cardiologist-level performance in arrhythmia diagnosis²⁰. However, these models generally require extensive labeled data for supervised learning and are typically task-specific^{1,20}, limiting their adaptability to other tasks or new demands¹⁷.

In recent years, the paradigm of AI has shifted from traditional deep learning to the emergence

of large language models (LLMs). Unlike supervised learning, LLMs, such as generative pre-trained transformers (GPT)²¹ and LLaMA²², are generative models pre-trained on vast amounts of unlabeled data through self-supervised learning. These models, typically comprising billions of parameters, excel in language processing and adapt to various tasks. To date, LLMs have demonstrated superior performance in clinical scenarios²³, including question answering (QA)²⁴, information retrieval²⁵, and clinical report generation^{26,27}. Recently, increasing numbers of studies have verified the effectiveness of LLMs for diagnostic tasks. For instance, PathChat²⁸, a vision-language generalist LLM fine-tuned on hundreds of thousands of instructions, achieved state-of-the-art performance in human pathology. Med-MLLM²⁷, a multimodal LLM pre-trained and fine-tuned on extensive medical data, including chest X-rays, CT scans, and clinical notes, demonstrated notable accuracy in COVID-19 diagnosis. Additionally, Kim et al.²⁹ employed GPT-4 with prompt engineering and found it surpassed mental health professionals in identifying obsessive-compulsive disorder, which underscores LLM's potential in mental health diagnostics.

Although this research field has drawn wide attention, many key questions remain under-explored. For instance, which diseases and medical data have been investigated in LLM-based diagnostic tasks (Q1)? What LLM techniques have been applied to disease diagnosis and how to choose appropriate ones (Q2)? What evaluation methods are appropriate for assessing performance (Q3)? Despite numerous review papers have investigated the studies of applying LLMs in medicine domain³⁰⁻³⁷, these efforts typically provide a broad overview of various clinical applications without underscoring disease diagnosis. For instance, Pressman et al.³⁸ offered a comprehensive summary of potential clinical applications of LLMs, including pre-consultation, treatment, postoperative management, discharge, and patient education. Additionally, none of these review papers address the nuances and challenges of applying LLMs to disease diagnosis or answer the

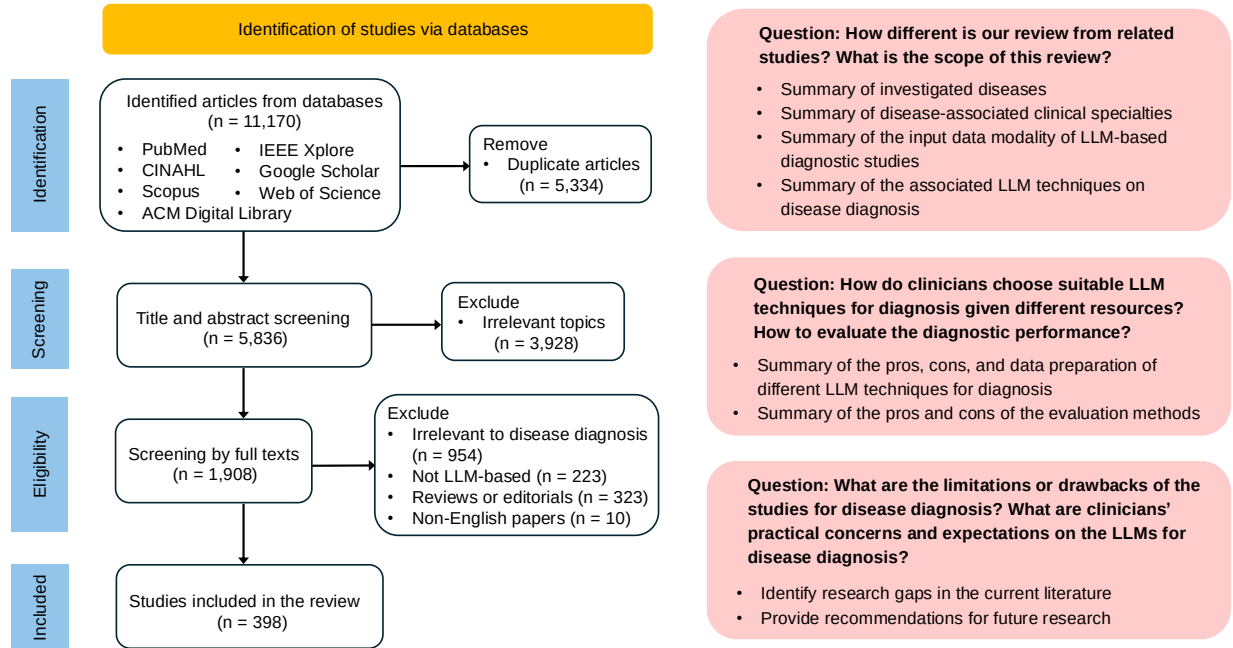


Fig 1 PRISMA flowchart of study records. PRISMA flowchart showing the study selection process.

forementioned questions, highlighting a critical research gap.

The primary aim of our review is to provide an overview of studies utilizing LLMs for disease diagnosis. The review introduced various disease types, disease-associated clinical specialties, clinical data, LLM techniques, and evaluation methods from existing works. Additionally, we provided recommendations for data preparation, selecting appropriate LLM techniques, and employing suitable evaluation strategies for diagnostic tasks. Further, our review characterized the limitations of current studies and shed insight into the challenges and future directions in this field. To the best of our knowledge, it is the first review that focused on disease diagnosis with LLMs and provided a comprehensive overview of this domain. In summary, this review outlined a blueprint for LLM-based disease diagnosis and helped to inspire and streamline future research efforts.

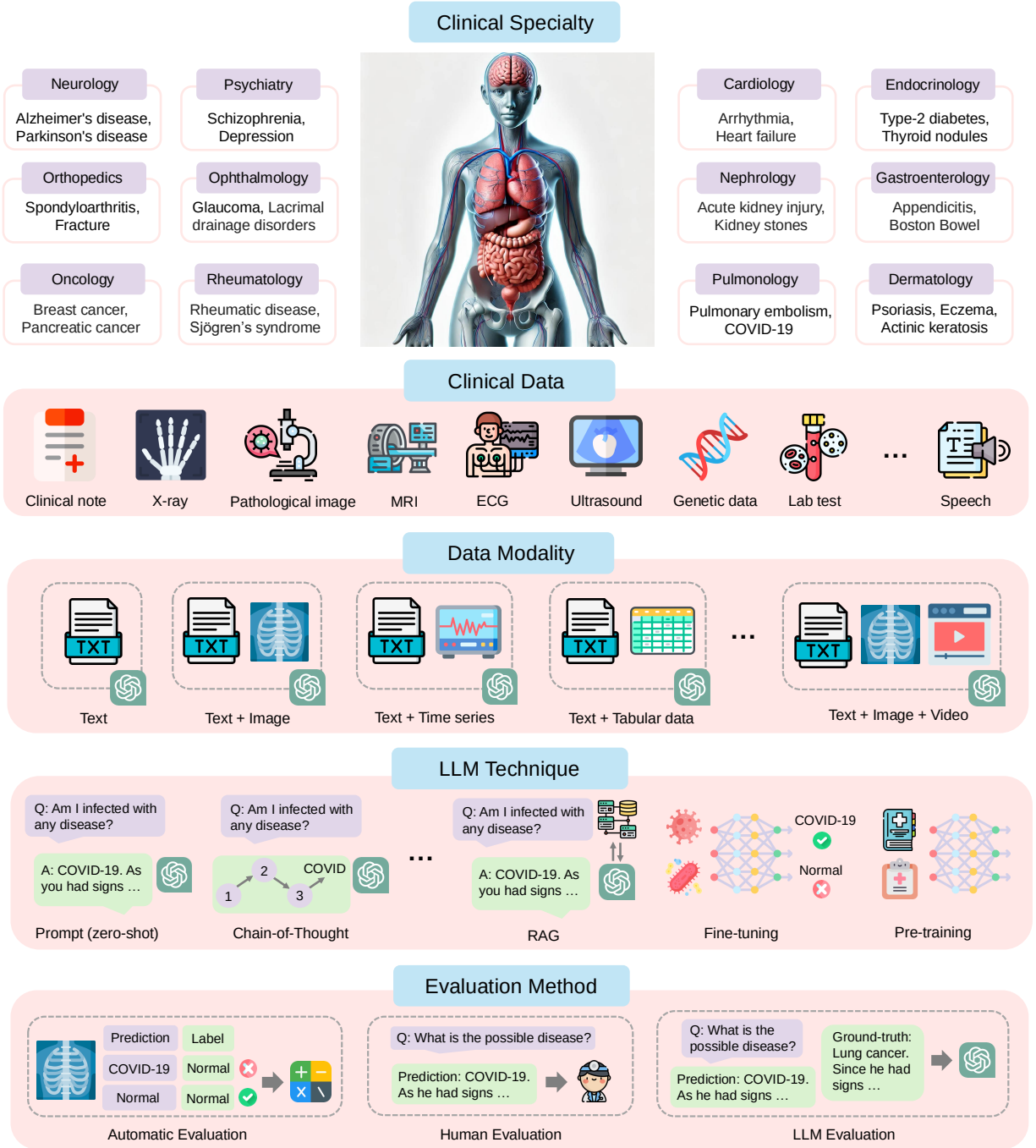


Fig 2 Overview of the investigated scope. It illustrated disease types and the associated clinical specialties, clinical data types, modalities of the utilized data, the applied LLM techniques, and evaluation methods. We only presented part of the clinical specialties and some representative diseases.

Table 1 Overview of LLM techniques for disease diagnosis.

Techniques	Types	Characteristics	Representative studies
Prompting	Zero-shot	A single instruction describing the task	Text ^{39,40} , image ^{41,42} , audio ^{43,44} , text-image ⁴⁵ , text-time series ^{46,47} , text-tabular ⁴⁸
	Few-shot	An instruction supplemented with several demonstrations	Text ^{49,50} , image ⁵¹ , text-image ^{52,53} , text-image-tabular ⁵⁴
	CoT	Decomposes a problem into multiple linear steps	Text ^{55,56} , audio ⁵⁷ , time series ⁵⁸ , text-image ^{59,60}
	Self-consistency	Generates multiple reasoning paths	Text ⁶¹ , audio ⁶² , text-image-tabular-time series ⁶³
	Soft prompt	Continuous vector embeddings with learnability	Text ⁶⁴ , image ⁶⁵ , tabular-time series ^{66,67} , text-image-graph ⁶⁸
RAG	Knowledge graph	External knowledge is stored in graphical structure	Text ⁶⁹⁻⁷¹ , text-time series ⁷²
	Corpus	External knowledge comes from high-quality corpora	Text ^{73,74} , text-image ^{75,76} , text-time series ⁷⁷
	Database	External medical knowledge comes from databases	Text ⁷⁸⁻⁸⁰ , text-image ^{81,82} , text-time series ⁸³
Fine-tuning	SFT	Injects medical knowledge via supervised learning	Text ⁸⁴⁻⁸⁶ , text-image ⁸⁷⁻⁸⁹ , text-video ^{90,91} , text-audio ^{92,93} , text-tabular ^{48,94,95}
	RLHF	Aligns the model with human preferences	Text ⁹⁶⁻⁹⁸ , text-image ⁹⁹
	PEFT	Fine-tunes a small number of (extra) model parameters	Text ^{84,100,101} , text-image ¹⁰²
Pre-training	-	Learns general knowledge with unsupervised learning	Text ^{101,103,104} , text-image ^{88,105,106} , text-tabular ^{48,107} , text-video ⁹³ , text-omics ¹⁰⁶

Note: SFT = supervised fine-tuning, RLHF = reinforcement learning from human feedback, PEFT = parameter-efficient fine-tuning.

Results

Overview of the scope

This section presented the scope of our review. Figure 2 not only illustrated disease types, the associated clinical specialties, clinical data types, and data modalities (Q1) but also introduced the applied LLM techniques (Q2) and evaluation methods (Q3), which answered the aforementioned questions. Specifically, we investigated 19 clinical specialties and over 15 types of clinical data in disease diagnosis. The clinical data spanned various data modalities, including text, image, video, audio, time series, and multimodal cases. Besides, we categorized existing works for disease diagnosis based on the applied LLM techniques, such as prompt (zero-shot), retrieval-augmented generation (RAG), and pre-training. Table 1 summarized the taxonomy of the mainstream LLM techniques. Figure 4 showcased the association of clinical specialties, data modalities, and the LLM techniques of the included papers. The above figures comprehensively revealed the current development of LLM-based disease diagnosis. Additionally, Figure 3 showed the meta-information analysis of our review, involving publication tendencies of different regions, a summary of widely-used LLMs for training and inference, and the statistics of data sources, evaluation methods, and data privacy status.

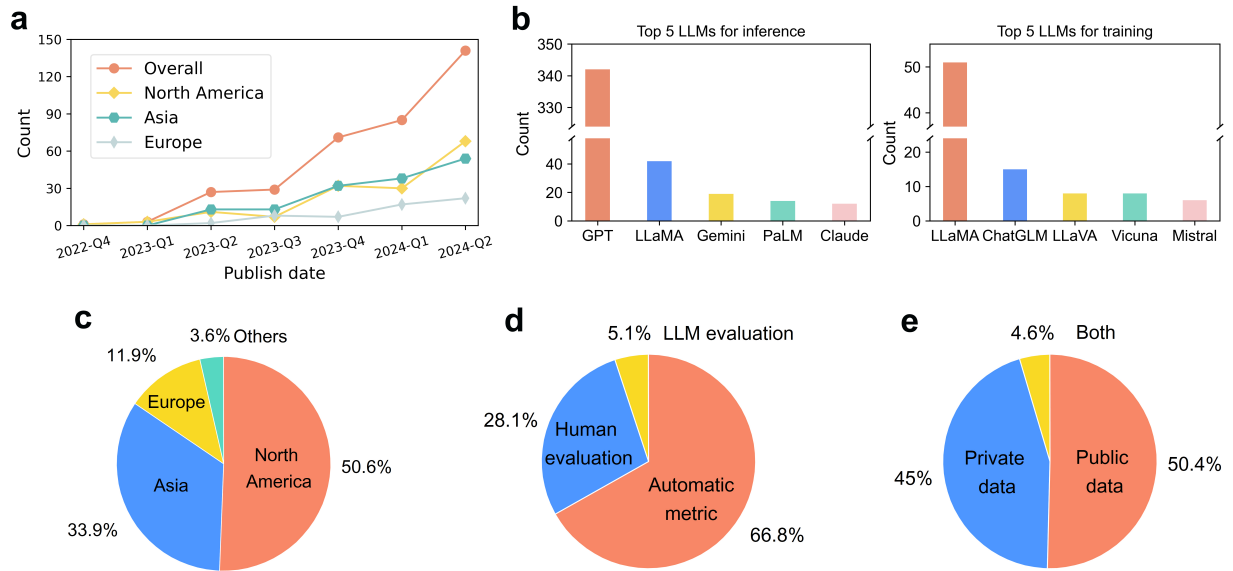


Fig 3 Metadata of information from LLM-based diagnostic studies in the scoping review. **a** Quarterly breakdown of LLM-based diagnostic studies. Since the information for 2024-Q3 is incomplete, our statistics only cover up to 2024-Q2. **b** The top 5 widely-used LLMs for inference and training. **c** Breakdown of the data source by regions. **d** Breakdown of evaluation methods (note some papers utilized multiple evaluation methods). **e** Breakdown of the employed datasets by privacy status.

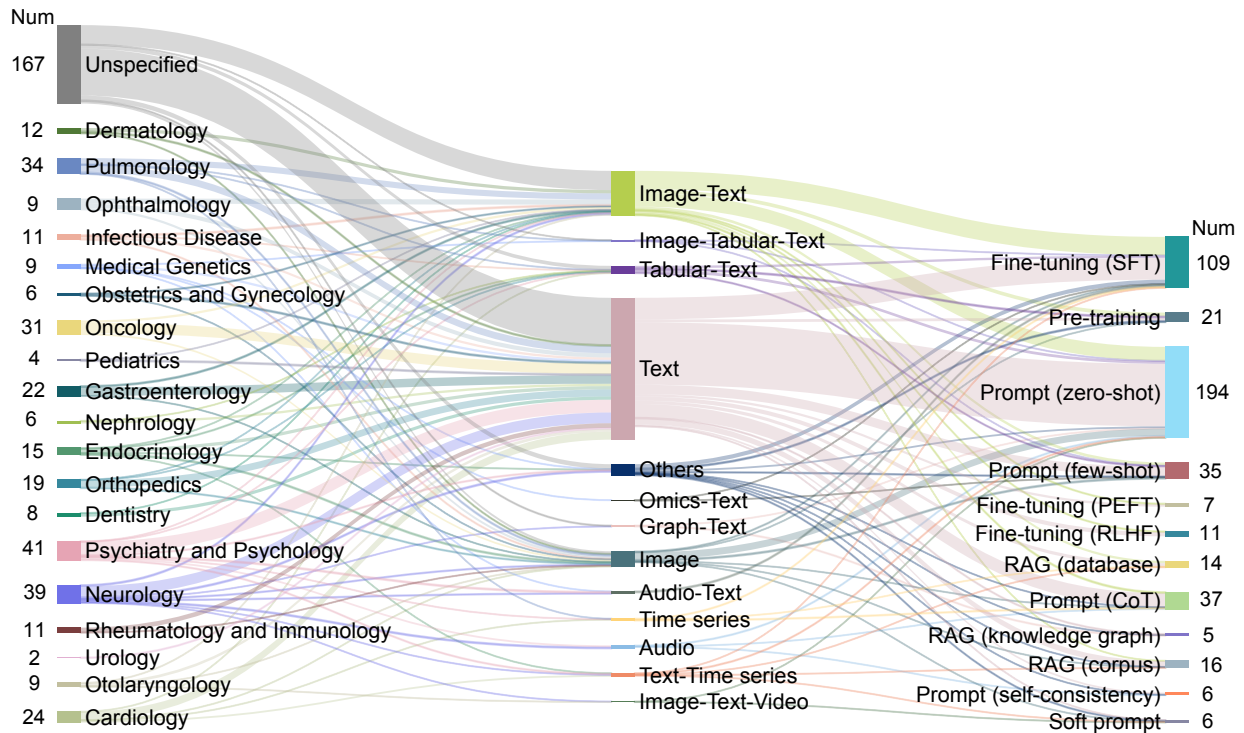


Fig 4 Summary of the association between clinical specialties (left), data modalities (middle), and LLM techniques (right) across the included papers.

Prompt-based disease diagnosis

A customized prompt typically comprises four components¹⁰⁸: instruction (specifying the task), context (defining the scenario or domain), input data (identifying the data to be processed), and output indicators (directing the model on the desired style or role). Over 60% (N=278) of included studies were prompt-based methods. We identified five distinct techniques that fall into two primary categories: hard prompts and soft prompts. Hard prompts include methods such as zero-shot, few-shot, Chain-of-Thought (CoT), and self-consistency prompting. These prompts are static and interpretable, written in natural language, which makes them particularly effective when the input and output structures are well-defined¹⁰⁹. On the other hand, soft prompts are continuous vector embeddings generated by a small, trainable model and then fed into an LLM. This technique, known as prompt tuning, encodes input data into task-specific embeddings, enabling the LLM to adapt to the task more effectively¹¹⁰.

Among the prompt-based studies, zero-shot prompting, which consists of a single instruction without labeled examples, was the most prevalent (N=194). CoT-based methods (N=37) were featured by breaking down complex problems into smaller, manageable parts, allowing the model to address these sequentially in multiple steps^{111,112}. For instance, in differential diagnoses, LLMs using CoT reasoning can follow clinical guidelines to sequentially interpret medical images, radiology reports, and symptom descriptions, providing intermediate outputs at each step that feed into subsequent analyses^{55,59,60}. This step-by-step approach allows the model to integrate context throughout the reasoning process, ultimately enabling a holistic final diagnosis. Few-shot prompt-based methods (N=35) expanded zero-shot prompting with a few labeled examples to enhance task performance. Studies based on self-consistency prompting (N=4) were characterized by generat-

ing multiple reasoning paths to enhance the reliability and robustness of LLMs^{63,113}. For example, Kim et al.⁶³ employed self-consistency prompting to predict depression scores (PHQ-4) by synthesizing diverse information from demographics, health domain literature, self-reported symptoms, and wearable sensor data to select the most consistent response among multiple reasoning paths. Soft prompt-based studies (N=6) involved training continuous vector embeddings before feeding them into LLMs, which enabled them to adapt LLMs' behavior for specific tasks. It has been mainly utilized to encode multimodal electronic health records (EHR), including medical images, clinical notes, and lab results. A key advantage of the soft prompt is its capacity to integrate external domain knowledge, such as medical concept embeddings, with contextual information like individual clinical profiles. This allows the model to generate nuanced disease diagnoses with detailed explanations, making it well-suited for complex clinical scenarios^{65,66}.

The majority of prompt-based studies involved unimodal data exploration (N=221), with most studies focusing exclusively on text data (N=171). Clinical text data such as clinical notes^{114,115}, medical imaging reports^{56,116,117}, and clinical case reports^{45,118} were predominantly utilized. These studies typically input clinical notes or case reports and ask LLMs for suggested disease diagnosis^{119–122}. Some studies (N=19) applied prompt engineering to medical image data. Commonly studied medical images included CT scans^{51,123}, X-rays^{68,124}, magnetic resonance imaging (MRI)^{51,125}, and pathological images^{126,127}. The primary use is to detect abnormalities on medical images and provide supporting evidence for differential diagnoses^{41,75,126,128}.

With the rapid development of multimodal LLMs, an increasing number of studies have explored using these models for disease diagnosis with prompt engineering (N=57). A key advancement in this area is visual-language models (VLMs) (e.g., GPT-4V, LLaVA, and Flamingo), which have made image-text pairs the most prevalent input combinations for multimodal LLMs (N=37).

Differing from the unimodal LLMs, VLMs were given more comprehensive clinical profiles, i.e., medical images and complementary textual descriptions, and were able to justify the diagnosis decisions with more details^{129–131}. For instance, Upadhyaya et al.⁷⁵ demonstrated that incorporating ophthalmologist feedback and contextual information (e.g., image location, purpose) with eye movement images significantly enhanced GPT-4V’s diagnostic accuracy for amblyopia.

More advanced multimodal LLMs, such as GPT-4o and Gemini-1.5 Pro, enabled prompt-based research to extend beyond text and image and include diverse data modalities for disease diagnosis. Specifically, many efforts leveraged audio and video data to facilitate the diagnosis of neurological and neurodegenerative disorders, such as autism^{43,132} and dementia^{44,68}. Some studies investigated using omics data for the detection of rare genetic disorders¹³³ and Alzheimer’s disease¹³⁴. Additionally, a wide range of risk prediction tasks tended to incorporate multimodal data for early warning, including time series data, such as ECG signals^{46,47,135} and wearable sensor data^{58,63}; tabular data, such as user demographics^{134,136}, and lab test results^{66,137}. The applications included depression and anxiety screening⁶³, emergency triage¹³⁸, and arrhythmia detection^{46,135,139}. Another study further combined multimodal LLMs with a medical concept graph for neurological disorder diagnosis⁶⁸.

Retrieval-augmented LLMs for diagnosis

To enhance the accuracy and credibility of the diagnosis, alleviate hallucination issues and update LLMs’ stored medical knowledge without needing re-training, recent studies^{69,70,79,140–142} have incorporated external medical knowledge into diagnostic tasks. The external knowledge primarily comes from corpus^{73,74,74–77,140,141,143–148}, databases^{61,78–83,123,135,142,149–153}, and knowledge graph^{69–72,154}, in the included papers. Based on the data modality, these RAG-based studies can be

roughly categorized into text-based, text-image-based, and time-series-based augmentations.

In text-based RAG, the majority of research^{74,78,79,140,142,143,145,148,149,151–153} has adopted a basic retrieval strategy. In this approach, external knowledge is encoded into vector representations using sentence transformers (e.g., OpenAI’s text-embedding-ada-002), which serve as retrieval sources. Queries were similarly encoded, allowing the system to identify and fetch the most relevant knowledge by calculating the similarity between query vectors and source vectors. This combined information was then fed into LLMs using specially designed prompts to generate diagnostic results. However, two papers employed LLMs to search similar medical cases from the given content^{144,146}. Zhenzhu et al.¹⁴⁴ designed guideline-based GPT-agents to summarize and retrieve content for traumatic brain injury rehabilitation-related questions. McInerney et al.¹⁴⁶ utilized the LLM to extract evidence fragments from previous notes for evaluating the risk factors for cancer, pneumonia, and pulmonary edema. Four studies retrieved relevant content from knowledge graphs^{69–71,147,154}. One study leveraged regular expressions to match useful knowledge for pulmonary hypertension diagnosis¹⁴¹. Different from previous studies where only one LLM was utilized for diagnosis, Wang et al.⁸⁰ employed several LLMs, each of which was equipped with specific medical knowledge, for joint diagnosis.

In text-image data processing, a common approach^{75,81,82,123,151} involves extracting features from input images, converting these features into textual descriptions, and subsequently applying text-based enhancement techniques. For instance, Ferber et al.¹⁵¹ employed advanced models like GPT-4V to extract critical information from images to facilitate the retrieval of relevant documents in oncology diagnosis. Similarly, Ranjit et al.⁷³ employed multimodal models to directly compute similarities between image and text features for document retrieval. Notably, two studies fine-tuned LLMs using the retrieved documents to enhance diagnostic accuracy^{76,150}.

For time-series RAG, most studies focused on the electrocardiogram (ECG) analysis^{77,83,135}. For instance, Yu et al.⁷⁷ converted fundamental ECG conditions into improved text descriptions by utilizing the retrieved relevant information. Yu et al.¹³⁵ constructed a local database with specific domain knowledge for diagnosing arrhythmia and sleep apnea. Chen et al.⁸³ pretrained a model with a public ECG-Report dataset and fine-tuned the model for hypertension and myocardial infarction diagnosis. One study utilized the RAG method for readmission prediction based on multimodal EHR⁷².

Fine-tuning LLMs for diagnosis

Fine-tuning a LLM typically encompasses two pivotal stages: supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF). During the SFT stage, the model is trained on task-specific instruction-response pairs, enabling it to interpret instructions and generate responses across diverse modalities. This phase is crucial for establishing a foundational understanding of the model, facilitating the processing of inputs to produce desired outputs. Subsequently, the RLHF phase further refines the model by aligning its behavior with human preferences. Utilizing reinforcement learning, the model is optimized to generate responses that are more helpful, truthful, and congruent with human values¹⁵⁵, thereby ensuring compliance with societal expectations for ethical and effective AI.

Medical SFT enhances the in-context learning, reasoning, planning, and role-playing capabilities of LLMs, leading to improved diagnostic performance. During this process, inputs from various data modalities are integrated into the LLM’s word embedding space. Following the approach outlined in LLaVA¹⁵⁶, visual information is first converted into visual token embeddings using an image encoder and a projector. These embeddings, which match the dimensionality of lan-

guage token embeddings, are then fed into the LLM for end-to-end training. In this review, many studies focused on conducting SFT on medical texts for diagnostic purposes (N=49). The medical texts can be clinical notes^{84,95,157}, clinical QA pairs^{84,104,158–160}, medical dialogues^{100,161–164}, or medical reports^{90,102,165–167}. Lot of studies combined both medical texts and images to enhance disease diagnosis (N=43), such as X-ray images^{90,165,168–170}, MRI images^{102,170,171}, or pathology images^{92,106,172}. A few studies also explored the detection of diseases from medical videos^{90,91}, where video frames were sampled and transformed into visual token embeddings. To perform SFT effectively, it is crucial to collect high-quality responses to task-specific instructions. These instructions should be well-defined and diverse, covering a wide range of scenarios to ensure comprehensive training.

RLHF methods could be divided into two categories: online and offline. Online RLHF, a key process for the success of ChatGPT¹⁷³, first fits a reward model to datasets of prompts and human preferences over responses, then uses reinforcement learning algorithms like PPO¹⁷⁴ to update the LLM to maximize the learned reward model. Some explorations showed online RLHF could effectively improve the diagnostic ability of medical LLMs^{97–99}. For example, Zhang et al.⁹⁸ aligned their model with the characteristics of doctors and achieved robust performance on a wide range of medical QA tasks, including condition diagnosis and etiological analysis. However, the overall performance of online RLHF highly relies on the quality of the reward model, which is expected to give accurate rewards to LLM responses, and several works demonstrated that the reward model could suffer from issues like over-optimization¹⁷⁵ and shifting from initial data distribution¹⁷⁶. Meanwhile, the training process for reinforcement learning is often characterized by instability and challenges in control¹⁷⁷. Offline RLHF methods like DPO¹⁷⁸ cast RLHF as optimizing a simple classification loss, eliminating the need for a reward model. These methods are

also more stable and computationally lightweight and have proven useful in medical LLMs alignment^{96,101,179}. Yang et al.¹⁰¹ found that if the offline RLHF phase is removed, their model exhibited significant performance drops in doctor evaluations on pediatric benchmarks. To conduct RLHF, a high-quality dataset of prompts and responses with human preferences is crucial to train a well-calibrated reward model¹⁸⁰ for online RLHF or ensure the better convergence of DPO like offline RLHF algorithms¹⁸¹, whether from human experts¹⁷³ or powerful AI models¹⁸².

As the size of LLMs increases, their capabilities are correspondingly enhanced. Consequently, larger models are often preferred to ensure a robust foundational capacity for adaptation to downstream tasks. However, scaling up model size renders full training increasingly impractical, as it demands extensive GPU resources. Parameter-efficient fine-tuning (PEFT) offers a solution to this challenge by minimizing the number of parameters requiring fine-tuning. The most popular PEFT method is Low-Rank Adaptation (LoRA)¹⁸³, which introduces trainable rank decomposition matrices into each layer without modifying the model's architecture. LoRA is particularly favored due to its advantage of not adding inference latency. In this review, all the PEFT-based studies (N=7) used LoRA to reduce the training cost^{84,100–102,184–186}.

Pre-training LLMs for diagnosis

LLMs are initially pre-trained on extensive text corpora to perform next-token prediction. During this phase, the model learns the structure of language and acquires a vast amount of knowledge about the world. When pre-trained on medical texts, LLMs gain foundational medical knowledge, which proves valuable when adapting them for various downstream medical tasks, including medical diagnosis. In this review, five studies perform text-only pretraining on the LLMs from different sources^{103,104,187–189}, such as clinical notes, medical QA texts, dialogues, and Wikipedia.

Table 2 Overview of evaluation metrics for disease diagnosis

Type	Evaluation metric	Purpose
Automatic metric	Accuracy ¹⁵⁸	The ratio of all correct predictions to the total predictions
	Precision ⁹⁵	The ratio of true positives to the total number of positive predictions
	Recall ⁹⁵	The ratio of true positives to the total number of actual positive cases
	F1 ²⁷	Calculated as the harmonic mean of precision and recall
	AUC ¹⁹²	The area under the Receiver Operating Characteristic curve
	AUPR ¹⁹³	The area under the precision-recall curve
	Top-k accuracy ¹⁹⁴	The ratio of instances with the true label in the top k predictions to total instances
	Top-k precision ¹²⁴	The ratio of true positives to total positive predictions within the top k predictions
	Top-k recall ¹⁹⁵	The ratio of true positives within the top k predictions to actual positive cases
	Mean square error ¹⁹⁶	The average of the squared differences between predicted and actual values
	Mean absolute error ¹¹⁴	The average of the absolute differences between predicted and actual values
	Cohen's κ ¹⁹⁷	Measure the agreement between predicted score and actual score
	BLUE ¹⁹⁸	Calculate precision by counting matching n-grams between reference and generated text
	ROUGE ⁴⁹	Calculate F1-score by matching n-grams between reference and generated text
	CIDEr ¹⁹⁹	Evaluate n-gram similarity, emphasizing alignment across multiple reference texts
Human evaluation	BERTScore ²⁰⁰	Measure similarity by comparing embeddings of reference and generated text
	METEOR ²⁰¹	Evaluate text similarity by considering precision, recall, word order, and synonym matches
	Necessity ⁴⁹	Whether the response or prediction assists in advancing the diagnosis
	Acceptance ²⁰²	The degree of acceptance of the response without any revision
	Reliability ²⁰³	The trustworthiness of the evidence in the response or prediction
Human or LLM evaluation	Explainability ¹⁴⁴	Whether the response or prediction is explainable
	Correctness ²⁰⁴	Whether the response or prediction is medically correct
	Consistency ²⁰⁵	Whether the response or prediction is consistent with the ground-truth or input
	Clarity ⁷⁹	Whether the response or prediction is clearly clarified
	Professionalism ²⁰³	The rationality of the evidence based on domain knowledge
	Completeness ⁴⁹	Whether the response or prediction is sufficient and comprehensive
	Satisfaction ²⁰⁶	Whether the response or prediction is satisfying
	Hallucination ²⁰⁵	Response contains inconsistent or unmentioned information with previous context
	Relevance ⁷⁹	Whether the response or prediction is relevant to the context
	Coherence ²⁰⁷	Assess logical consistency with the dialogue history

Moreover, eight studies injected medical visual knowledge into multimodal LLMs via pretraining^{88,105–107,189–191}. For instance, Chen et al.¹⁰⁵ and Wang et al.¹⁸⁹ pre-trained their models on visual question-answering (VQA) data. Specifically, Chen et al.¹⁰⁵ employed an off-the-shelf multimodal LLM to reformat image-text pairs from PubMed into VQA data points for training their model. To improve the quality of the image encoder, pretraining tasks like reconstructing images at tile-level or slide-level¹⁰⁶, and aligning similar images or image-text pairs⁸⁸ are common choices.

Evaluation strategy

As evaluating diagnostic performance is crucial, we further summarized and analyzed the evaluation strategies for diagnostic tasks. Generally, existing evaluation methods fall into three categories: automatic evaluation, human evaluation, and LLM evaluation (shown in Table 2). An overview of the advantages and limitations of the evaluation strategies is depicted in Figure 5.

Most studies assessed diagnostic effectiveness using automatic metrics, which can be broadly categorized into three types. The first type primarily uses classification-based metrics such as accuracy, precision, and recall, which are suitable for single-disease prediction. For example, Liu et al.²⁷ adopted AUC, accuracy, and F1 score to evaluate COVID-19 diagnosis effectiveness. The second type is generally used in multi-label scenarios, where predictions involve multiple potential diagnoses, including top-k accuracy and top-k precision. For instance, Tu et al.¹⁹⁴ utilized top-k accuracy to measure the percentage of correct diagnoses appearing within the top-k positions of the diagnosis list. The third type applies to risk prediction tasks, where mean absolute error (MAE) or mean squared error (MSE) measures the deviation between predicted values and the actual ones^{114,196}. In summary, automatic metrics offer advantages such as time and cost efficiency, ease of implementation, and suitability for large-scale data. However, they require ground-truth answers, which are often unavailable in many scenarios. Additionally, these metrics typically lack human-centric perspectives, such as assessing the reliability or overall usefulness of the prediction. Furthermore, they generally fall short in evaluating complex scenarios, such as determining whether a diagnostic reasoning process is medically correct²⁰⁸.

Many studies evaluated diagnostic performance through human efforts^{24,209}. This method relies on domain experts to evaluate the quality of model predictions based on their medical knowledge. One advantage lies in that it typically does not require ground-truth answers. Additionally, it accommodates human-centric perspectives and can address complex tasks that necessitate extensive human intelligence or domain knowledge. However, human evaluation presents several limitations, including significant time and cost demands, as well as a susceptibility to human error. Consequently, this strategy is usually applied for small-scale data assessment.

Additionally, some studies have utilized LLMs to replace human experts in diagnostic evalua-

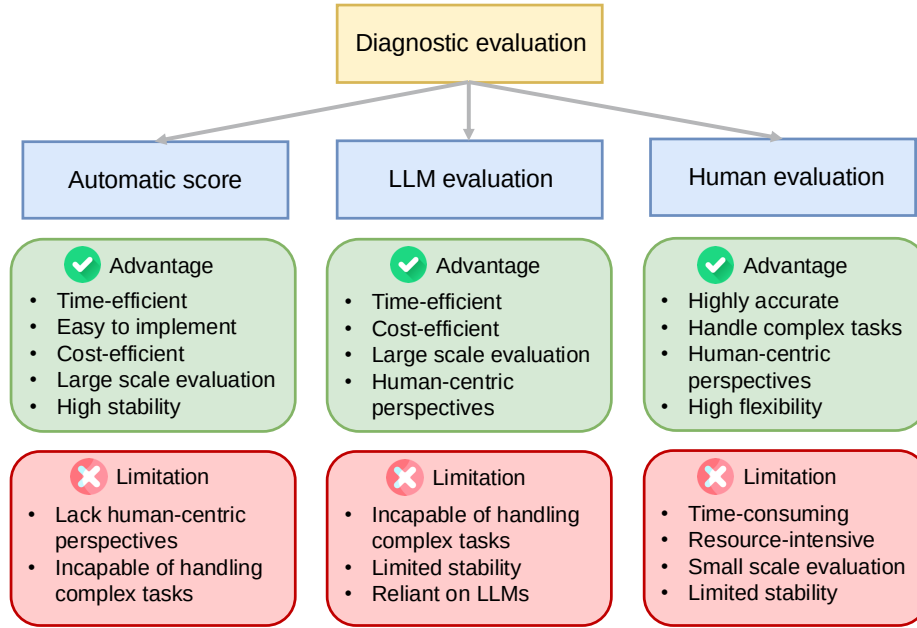


Fig 5 Summary of the evaluation strategies for diagnostic tasks.

tion^{210–212}. LLM evaluation combines the benefits of human-centric evaluation with the efficiency of automated metrics. Although ground-truth is not strictly required for this approach^{205,212}, its inclusion further enhances the reliability of LLM evaluation²⁰⁹. Commonly used LLMs for this purpose include GPT-3.5, GPT-4, and LLaMA-3. However, this approach is limited by the performance of the employed LLMs, which are susceptible to hallucination issues²⁰⁵. Moreover, LLM-based evaluation may struggle with handling complex clinical scenarios²¹³.

In summary, the above evaluation strategies have their advantages and limitations. The balance between accurate evaluation and cost-effectiveness varies depending on the specific scenario. Our analyses, presented in Figure 5, provide convenience in selecting appropriate evaluation strategies, catering to the requirements of various applications.

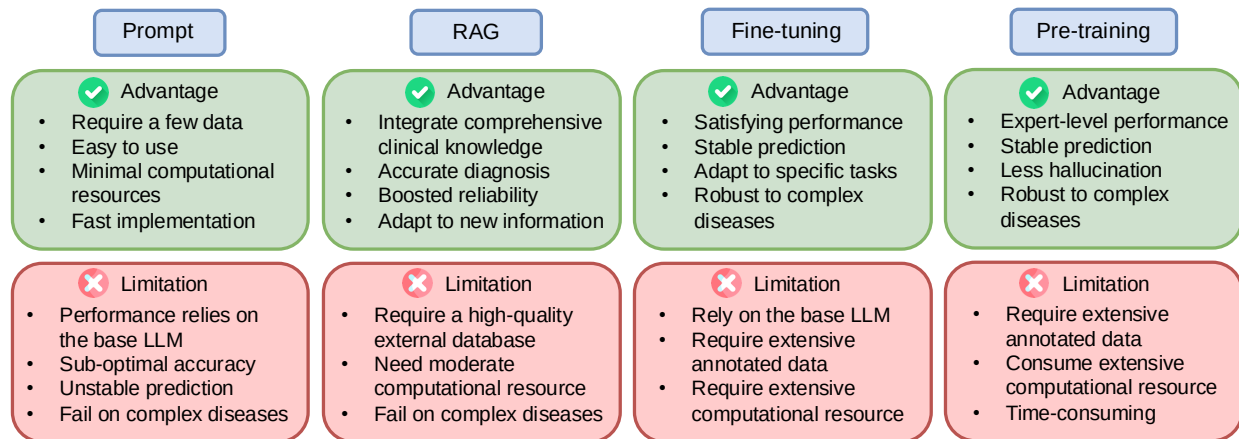


Fig 6 Summary of the advantages and limitations of the mainstream LLM techniques for diagnosis.

Discussion

This section presented notable findings from the included studies, discussed the data preparation for the mainstream LLM techniques, and highlighted key challenges and potential future research directions. Our review revealed that most studies utilized LLMs for disease diagnosis through prompt learning. The phenomenon might be explained as follows. Firstly, it requires minimal data. For instance, zero-shot and few-shot prompts enable the development of diagnostic systems with just a few dozen examples^{39,214}. Secondly, prompt-based methods are user-friendly and require minimal setup, making them accessible to researchers with limited machine-learning expertise. Additionally, it significantly reduces computational overhead, making implementation feasible on ordinary hardware. Furthermore, when used appropriately, large-scale LLMs like GPT-4 or GPT-3.5, which own extensive medical knowledge, demonstrate fair performance across various diagnostic tasks^{24,214}.

We summarized the advantages and limitations of mainstream LLM techniques of the included papers in Figure 6 and discussed the data preparation as follows. Generally, the selection of LLM techniques for developing diagnostic systems depends on the quantity and quality of available data.

Specifically, prompt engineering is highly flexible and effective when annotated data is limited. Generally, designing an appropriate instruction supplemented with several examples as demonstration is sufficient for prompting²⁴. Zero-shot prompting even allows models to perform diagnosis without annotated examples while still achieve fair performance²¹⁴. To effectively apply RAG to diagnosis, a comprehensive and high-quality external knowledge base is indispensable. This knowledge base can be databases⁷⁹, corpora^{78,143} or knowledge graphs⁷⁰ from which LLMs can retrieve accurate information during inference. Effective fine-tuning necessitates a well-annotated, domain-specific dataset that includes labeled examples reflecting the target diagnostic tasks, such as annotated clinical notes or medical images, and a substantial number of samples²⁷. Pre-training requires extensive and diverse datasets that encompass a wide spectrum of medical knowledge, including unstructured text (e.g., clinical notes, medical literature) or structured data (e.g., lab test results)^{54,94}. The quality and diversity of the pre-training datasets are crucial for establishing the model's foundational knowledge and its ability to generalize across various medical contexts. While pre-training and fine-tuning would achieve promising performance and reliability^{27,190}, they demand significant resources, such as advanced graphics cards and millions of medical data, which are usually hard to obtain. In contrast, not all scenarios require expert-level performance for disease diagnosis, such as large-scale screening^{8,215}, health risk alerts from mobile devices⁵⁸, or public health education^{30,32}. Balancing the trade-off between accuracy and cost-effectiveness varies by scenario. In summary, the analyses presented in Figure 6 guide users in selecting appropriate LLM techniques for disease diagnosis based on available resources.

Despite the progress in LLM-based methods for disease diagnosis, this scoping review identifies several barriers that impede their clinical utility (Figure 7). In the information-gathering process, a notable limitation is that only a small subset of studies integrated comprehensive mul-

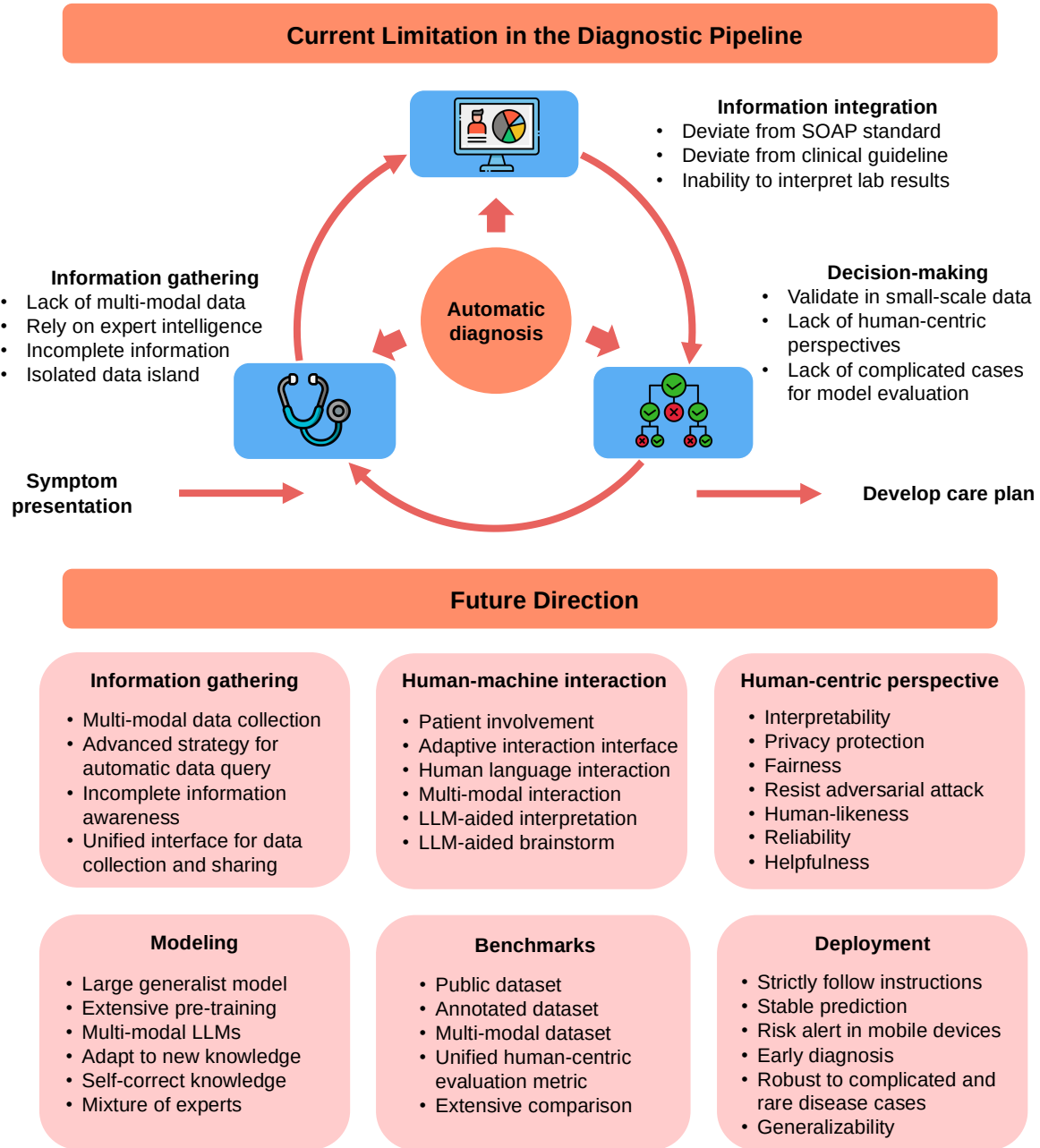


Fig 7 Summary of the limitation and future direction for LLM-based disease diagnosis.

timodal data for diagnosis²¹⁶, such as text, image, time series, and other modalities. For example, Deng et al.²¹⁷ developed a multimodal LLM incorporating text, images, video, and speech for autism spectrum disorder screening. This discrepancy contrasts with real-world diagnostic scenarios, where comprehensive patient information spans multiple data modalities¹⁶⁰, particularly for complex conditions affecting multiple organs. Therefore, future research should emphasize collecting and fusing information from diverse modalities to simulate real-world scenarios.

Another limitation is that most studies implicitly assume the collected patient information is sufficient for disease diagnosis. Nevertheless, this assumption usually hardly holds, particularly in initial consultations or with complicated diseases, and using incomplete data would likely cause misdiagnosis^{218,219}. In practice, clinical information gathering is an iterative process, beginning with the collection of initial patient data (e.g., subjective symptoms), narrowing down potential diagnoses, and then conducting medical examinations for further data collection and disease screening²²⁰. This process typically requires extensive domain expertise from experienced clinicians. To alleviate the reliance on professionals, an increasing number of studies are exploring diagnostic conversations that collect relevant patient information through multi-round dialogues^{221,222}. For example, AIME utilized LLMs for clinical history-taking and diagnostic dialogue¹⁹⁴, while MEDIQ asked follow-up questions to gather essential information for clinical reasoning²¹³. Following this tendency, future research can integrate the awareness of incomplete information into diagnostic models or develop advanced methods for automatic diagnostic queries^{223,224}.

Some barriers lie in the information integration process. Although adhering to clinical guidelines is critical in medical scenarios, only a few studies considered this factor. For instance, Kresevic et al.¹⁴³ aimed to improve clinical decision support systems through accurate interpretation of medical guidelines for chronic Hepatitis C Virus infection management. Future works can integrate

clinical guidelines for developing diagnostic systems. Besides, the integration and interpretation of lab test results pose significant value in healthcare. For example, He et al.²²⁵ exploited LLMs to generate lab test-related responses to answer patients' queries, thus gaining patients' trust. A future direction is leveraging LLMs to interpret lab test results for professionals and patients.

Exploring the interaction between clinicians, patients, and diagnostic systems presents a promising avenue for research^{221,222,226}. In medical settings, diagnostic systems could function as assistants that provide supplementary information to enhance the accuracy or efficiency of clinicians^{51,157,227,228}. Besides, these systems should incorporate feedback from medical experts, facilitating continuous refinement and adaptation. Additionally, a user-friendly interface is expected for human-machine interaction. For instance, doctors directly talk with the diagnostic systems to input patients' information and perform discussions. In brief, future studies could explore how the effective application of diagnostic algorithms can further enhance clinical significance²²⁹.

Another barriers lie in the decision-making step. While many studies emphasize diagnostic accuracy, they usually ignore human-centric perspectives such as model interpretability, patient privacy, safety, and fairness^{30,230,231}. Specifically, providing diagnostic predictions alone is insufficient in clinical scenarios, as the black-box nature of LLMs often undermines trust^{205,208}. Accordingly, it is essential to provide interpretative insights into the diagnoses²⁰⁸. For example, Dual-Inf is a prompt-based framework that not only offers potential diagnoses but also explains the rationale behind them²⁰⁹. Regarding privacy, adherence to regulations like the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR) is essential, such as the de-identification of sensitive information^{25,232}. To date, only a few works have investigated the issue^{80,233}. For instance, SkinGPT-4 is a dermatology diagnostic system designed for local deployment to protect user privacy²³³. Fairness is another concern, ensuring patients are

not discriminated against based on gender, age, or race²³⁰. Research addressing the fairness issue in LLM-based diagnosis remains limited^{234,235}. In short, future research should integrate these human-centric perspectives into diagnostic systems to address these critical issues.

In terms of technical aspects, integrating multimodal data for disease diagnosis draws increasing attention¹². However, several challenges remain, including eliminating data noise²³⁶, fusing heterogeneous data from various modalities²³⁷, and performing efficient learning. Besides, many domain-specific LLMs are constrained by smaller parameter scales compared to general-domain LLMs^{203,238}. This may be due to the lack of substantial corpora and computational resources necessary for training large-scale medical models¹⁹⁴. However, pre-training on vast medical datasets can embed more medical knowledge into LLMs, thereby enhancing their reasoning abilities and improving performance on rare diseases and complex cases^{239,240}. Future work can also investigate employing multiple specialist models to boost diagnostic accuracy, as it simulates interdisciplinary clinical discussions for complex disease cases involving multiple clinical specialties^{80,241,242}. Additionally, hallucination is a long-standing issue in LLMs, which severely jeopardizes the reliability of diagnostic systems²⁴³. To mitigate data-related hallucination, which is rooted in the misinformation or knowledge gap from training data, future studies can investigate knowledge editing²⁴⁴ or retrieve external knowledge^{79,143} for diagnosis. For the training-related hallucinations that are raised by the intrinsic limitations of the architecture or training strategies in LLMs²⁴⁵, future works can explore novel model architectures or pre-training strategies^{239,246}.

Another critical area is the development of diagnostic systems. Many studies utilized private datasets, which are often inaccessible due to privacy concerns^{143,247}. However, the advancement of diagnostic systems necessitates a greater availability of public data. The other issue is that the scarcity of annotated data poses a significant challenge to the development of this field. This is be-

cause well-annotated datasets enable exploiting automatic metrics for evaluation, reducing the need for extensive human effort in performance assessment²⁰⁹. Therefore, constructing and releasing annotated benchmark datasets would significantly contribute to the research community. Moreover, performance evaluation should also be highlighted. Currently, there is no standardized guideline for evaluating diagnostic performance, particularly regarding human-centric metrics^{49,207,248}. A generic principle is to consider metrics from different aspects, such as effectiveness, robustness, reliability, and explainability, thereby providing a comprehensive evaluation.

In practice, the deployment of diagnostic systems remains a considerable challenge. Many studies reported that LLMs struggle to provide stable responses or predictions^{231,249}. For instance, Hager et al.²³¹ discovered that the changes in instructions could result in large obvious changes in diagnostic accuracy. However, a stable and reproducible clinical decision is crucial in clinical scenarios. Therefore, future works can explore ensuring the stability of LLMs for diagnostic tasks. The other direction is to deploy diagnostic algorithms on mobile devices that can continuously and automatically collect basic signs and information from the human body, such as electroencephalogram rhythms and ECG rhythms. This enables mobile devices to send health-related risk alerts for early warning. In addition, early diagnosis draws wide attention and creates significant value^{16,237}. For instance, early diagnosis of lung adenocarcinoma can increase the 5-year survival rate to 52%²⁵⁰. However, only a few studies exploited LLMs for this purpose^{75,121}. The difficulty lies in that many diseases typically lack obvious symptoms in the early stages and are hard to identify. Future directions can further explore how to deploy diagnostic systems for early diagnosis.

In conclusion, our study provided a comprehensive review of LLM-based methods for disease diagnosis. Our contributions were multifaceted. First, we summarized the disease types, the associated clinical specialties, clinical data, the employed LLM techniques, and evaluation methods

within this research domain. Second, we compared the advantages and limitations of mainstream LLM techniques and evaluation methods, offering recommendations for developing diagnostic systems based on varying user demands. Third, we identified intriguing phenomena from the current studies and provided insights into their underlying causes. Lastly, we analyzed the current challenges and outlined the future directions of this research field. In summary, our review presented an in-depth analysis of LLM-based disease diagnosis, outlined its blueprint, inspired future research, and helped streamline efforts in developing diagnostic systems.

Methods

Search strategy and selection criteria

This scoping review is reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines, as shown in Figure 1. We performed literature search from various resources to find relevant articles published between 1 Jan 2019 and 18 July 2024. We searched seven electronic databases, including PubMed, CINAHL, Scopus, Web of Science, Google Scholar, ACM Digital Library, and IEEE Xplore. The search terms were selected based on consensus expert opinion and used for each database (see Supplementary Data 1).

We performed a two-stage screening process to focus on LLMs for human disease diagnosis. The first stage involved using the title and abstract for paper exclusion. The criterion was as follows: (a) articles were not published in English; (b) articles irrelevant to LLMs or foundation models; and (c) articles irrelevant to the health domain. The second stage was full-text screening, emphasizing using language models for diagnosis-related tasks. We excluded review papers, editorials, and papers not explicitly used for disease diagnosis. Notably, the scope of “disease diagnosis” in this review was not confined to tasks that directly produced diagnoses, such as medical

image classification; it also encompassed diagnosis-related tasks, such as depression identification⁸ and suicide risk prediction⁹. See Supplementary Data 2 for details of the scope. We also excluded studies concerning foundation models that do not incorporate text modalities, including visual foundation models. Full texts of studies reserved from the initial screening were independently evaluated for final eligibility by at least two examiners. Any disagreements were resolved by consensus or a third member.

Data extraction

Information garnered from the articles consists of four categories. (1) Basic information, including title, published venue, published time (year and month), and region of correspondence. (2) Data-related information, including data sources (continents), dataset type, modality (e.g., text, image, video, or text-image), clinical specialty, disease name, data availability (i.e., private or public data), and data size. (3) Model-related information, which comprises base LLM type, parameter size, and technique type. (4) Evaluation, which includes evaluation schema (e.g., automatic or human evaluation) and evaluation metric (e.g., accuracy and precision). See Supplementary Table 1 for details of the data extraction form.

Data synthesis

We synthesized insights from the data extraction to highlight the principal themes in LLM-based disease diagnosis. Firstly, we presented the scope of our review, spanning disease-associated clinical specialties, clinical data, data modalities, and LLM techniques. We also calculated the statistics of the meta-information, including development tendencies, the most widely used LLMs, and the distribution of the data sources. We then summarized various LLM-based techniques and eval-

uation strategies, analyzing their strengths and weaknesses, and offering targeted recommendations. Diving deeper into technical aspects, we detailed modeling approaches into four categories (prompt-based methods, RAG, fine-tuning, and pre-training), and fine-grained subtypes. We also examined the challenges faced by current research and outlined potential future directions. In summary, our synthesis encompassed a broad range of perspectives, assessing studies across data, LLM techniques, performance evaluation, and application scenarios, which are in line with established reporting standards.

Data availability

The analyzed data are included in this article. Aggregate data analyzed in this study will be released upon the acceptance of this paper.

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Author contributions

S.Z. conceptualized the study and led the work. Z.Z., S.Z., J.Y., and M.Z. searched papers. S.Z., Z.X., M.Z., C.X., Y.G., Z.Z., S.D., J.W., K.X., Y.F., L.X., and J.Y. conducted paper screening and data extraction. S.Z., Z.X., M.Z., and C.X. performed data synthesis and contributed to the writing. D.Z., G.M., and R.Z. revised the manuscript. R.Z. supervised the study. All authors read and approved the final version.

Competing interests

The authors declare no competing interests.